

1. Create internal and external customer tables and load data into them:

```
CREATE EXTERNAL TABLE customer_external (
  id INT,
  name STRING,
  city STRING,
  address STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
STORED AS TEXTFILE
LOCATION '/ext/cus/'

load data local inpath 'data/customers.csv'
into table customer_external

select * from customer_external
```

id	name	city	address
1	Ali	Sanaa	Street 10
2	Mohammed	Aden	"Building 5
3	Omar		Zone 3
4	Ahmed	Ibb	Old Street Behind School
5	Hassan	Sanaa	"Al-Zubairi St

```
CREATE TABLE customer_internal (
  id INT,
  name STRING,
  city STRING,
  address STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
STORED AS TEXTFILE

load data local inpath '/data/customers.csv'
into table customer_internal

select * from customer_internal
```

id	name	city	address
1	Ali	Sanaa	Street 10
2	Mohammed	Aden	"Building 5
3	Omar		Zone 3
4	Ahmed	Ibb	Old Street Behind School
5	Hassan	Sanaa	"Al-Zubairi St

2. -Notice the issue related to delimiter inside the address column value, find a solution to make hive read the file properly without changing the separator. (Do some web search):

The issue is hive think the address is two columns because there is a comma between the street number and state so since you asked to not change the separator I used OpenCSVSerde library because it give us the ability to tell hive to not deal with commas inside double quotations for example.

```
CREATE TABLE customer_internal (
  id INT,
  name STRING,
  city STRING,
  address STRING
)
ROW FORMAT SERDE 'org.apache.hadoop.hive.serde2.OpenCSVSerde'
WITH SERDEPROPERTIES ("separatorChar" = ",", "quoteChar" = "\"")
STORED AS TEXTFILE

load data local inpath 'itversity-material/data/customers.csv'
into table customer_internal

select * from customer_internal
```

id	name	city	address
1	Ali	Sanaa	Street 10
2	Mohammed	Aden	Building 5, City Center
3	Omar		Zone 3
4	Ahmed	Ibb	Old Street Behind School
5	Hassan	Sanaa	Al-Zubairi St, Opposite Bank

3. Try to drop both external and internal tables and notice what happens to the data.

In external table the metadata is removed but the data files are saved in hdfs.

In internal table both metadata and data are removed.

4. Create customer dimension (Slowly changing dimension type 2) as “customer_scd2_mixed.csv” file.

```
CREATE TABLE customer_scd2 (
  CustomerID INT,
  Name STRING,
  Email STRING,
  Phone_Number STRING,
  Address STRING,
  JOIN_Date DATE,
  Start_Date DATE,
  End_Date DATE,
  Is_Current INT
)
ROW FORMAT SERDE 'org.apache.hadoop.hive.serde2.OpenCSVSerde'
WITH SERDEPROPERTIES ("separatorChar" = ",", "quoteChar" = "\"")
STORED AS TEXTFILE
TBLPROPERTIES ('skip.header.line.count' = '1')

select * from customer_scd2
```

AZ customerid	AZ name	AZ email	AZ phone_number	AZ address	AZ join_date	AZ start_date	AZ end_date	AZ is_current
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5. Use “customer_updated.csv” file to insert new data with end_date = null and is_current = ‘1’ and update changed data to be with end_date and is_current = “0” to track history.

```
Load data local Inpath 'itversity-material/data/customer_scd2_mixed.csv'
into table customer_scd2

select * from customer_scd2
```

AZ customerid	AZ name	AZ email	AZ phone_number	AZ address	AZ join_date	AZ start_date	AZ end_date	AZ is_current
1	16324	Jennifer Harris	upd_michael77@example.n	001-438-988-5271x923	323 Yoder Place East Kath	2/3/2025	4/1/2025	1
2	13094	Lisa Morris	upd_brandtscoth@example.e	459.507.0988	85587 Cook Flats Jamesfur	3/6/2025	4/1/2025	1
3	13483	Jesus Cochran	upd_natalie29@example.co	001-911-846-8022	57000 Nancy Crescent Apt. 1	1/18/2025	4/1/2025	1
4	13819	Jacqueline Russo	upd_littlejamie@example.o	661.959.6302	827 Janet Port Apt. 218 Espl	1/9/2025	4/1/2025	1
5	16871	Curtis Smith	upd_carlos92@example.com	917.534.0110	4373 Gordon Tunnel Carmer	1/10/2025	4/1/2025	1
6	16201	Rachel Hall	upd_ppowers@example.org	3504375705	0650 Hall Extension Apt. 33v	1/13/2025	4/1/2025	1
7	16361	Jessica Alexander	upd_wilsonjustin@example.e	-4568	PSC 4826, Box 9054 APO AE	1/25/2025	4/1/2025	1
8	15184	Justin Parrish	upd_rachel20@example.org	747-565-9859	2617 Bush Squares Suite 03t	3/14/2025	4/1/2025	1
9	15197	Shannon Brown	upd_hullrussell@example.ne	-6905	99217 Julia Garden Robertvi	1/15/2025	4/1/2025	1
10	16221	David Vega	upd_averysheila@example.e	(651)570-7927x050	47486 John Stravenue South	1/20/2025	4/1/2025	1
11	15821	Melanie Smith DDS	upd_raiphelewis@example.n	-1947	7616 Kaitlyn Flats Apt. 553	2/1/2025	4/1/2025	1
12	17166	Nicholas Hall	upd_richardsonbruce@exam	001-936-283-3553x33522	4852 Joseph Village Meyers	1/20/2025	4/1/2025	1
13	13343	Nichole Buchanan	upd_martha94@example.ne	(987)496-2427x50241	92698 Gray Lakes South Edw	3/17/2025	4/1/2025	1
14	16318	Ashley Chang	upd_uquinn@example.com	512.415.4760x6146	8798 Lucas Valley Nicoletow	2/20/2025	4/1/2025	1
15	13299	William Massey	upd_brendacoleman@exam	338-306-9099x221	PSC 1878, Box 0974 APO AA	3/8/2025	4/1/2025	1
16	16618	Henry Wilson	upd_elizabeth01@example.e	676-753-4633	6594 Baker Brooks Apt. 716	3/26/2025	4/1/2025	1
17	15951	Jenny Miller	upd_brandonmercado@exa	+1-378-264-6903x15230	PSC 8968, Box 1943 APO AE	3/29/2025	4/1/2025	1
18	17317	Thomas Sosa	upd_johnsonlonnie@examp	802-359-0363	527 Jennifer Light Palmerbei	3/30/2025	4/1/2025	1
19	18120	Sean Mendoza	upd_gordonkeith@example	612-247-6675x190	4551 Davis Forest Apt. 998	3/1/2025	4/1/2025	1
20	12738	Jimmy Johnson	upd_candace51@example.n	(609)236-7497x6753	821 Harris Plains Tiffanyber	1/24/2025	4/1/2025	1

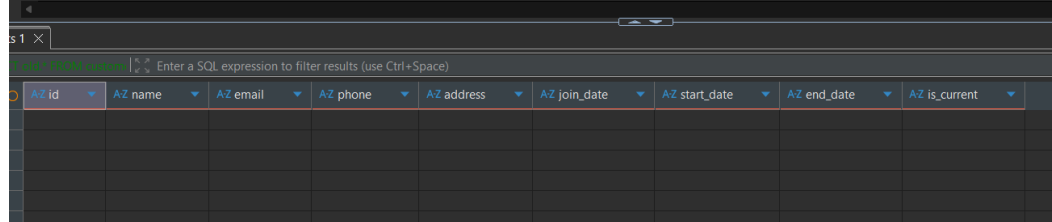
6. “Hive doesn’t support update or deletes so find a work around to make this SCD type 2 without using Transactional tables”

I created a new table for updated data and a temporary table for new data with scd2
So now I have three tables:

- customer_scd2: old data.
- customer_updates: new data.
- customer_scd2_new: old and new data.

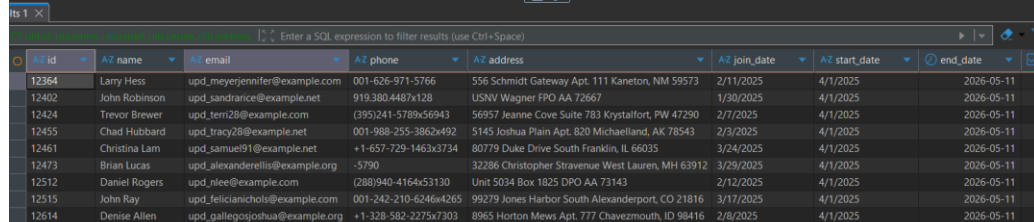
1. Inserting data that did not change to customer_scd2_new :

```
--customers with no changed data--
INSERT INTO TABLE customer_scd2_new
SELECT old.*
FROM customer_scd2 old
LEFT JOIN customer_updates upd
ON old.id = upd.id
WHERE (
    old.is_current = true
    AND old.name = upd.name
    AND old.email = upd.email
    AND old.phone = upd.phone
    AND old.address = upd.address
)
OR old.is_current = false;
```



2. Inserting old data of customers who have changed whit specifying end_date and is_current to FALSE:

```
--old records for customers with changed data--
INSERT INTO TABLE customer_scd2_new
SELECT
    old.id,
    old.name,
    old.email,
    old.phone,
    old.address,
    old.join_date,
    old.start_date,
    current_date() AS end_date,
    false AS is_current
FROM customer_scd2 old
JOIN customer_updates upd
ON old.id = upd.id
WHERE old.is_current = true
AND (
    old.name <> upd.name
    OR old.email <> upd.email
    OR old.phone <> upd.phone
    OR old.address <> upd.address
);
```



AZ id	AZ name	AZ email	AZ phone	AZ address	AZ join_date	AZ start_date	AZ end_date	AZ is_current
12364	Larry Hess	upd_meyerjennifer@example.com	001-626-971-5766	556 Schmidt Gateway Apt. 111 Kaneton, NM 59573	2/11/2025	4/1/2025	2026-05-11	
12402	John Robinson	upd_sandrance@example.net	919-380-4487x128	USNV Wagner FPO AA 72667	1/30/2025	4/1/2025	2026-05-11	
12424	Trevor Brewer	upd_terri28@example.com	(395)241-5789x56943	56957 Jeanne Cove Suite 783 Krystalfort, PW 47290	2/7/2025	4/1/2025	2026-05-11	
12455	Chad Hubbard	upd_tracy28@example.net	001-968-255-3862x492	5145 Joshua Plain Apt. 820 Michaelland, AK 78543	2/3/2025	4/1/2025	2026-05-11	
12461	Christina Lam	upd_samuel91@example.net	+1-657-729-1463x3734	80779 Duke Drive South Franklin, IL 66035	3/24/2025	4/1/2025	2026-05-11	
12473	Brian Lucas	upd_alexanderellis@example.org	-5790	32286 Christopher Stravenue West Lauren, MH 63912	3/29/2025	4/1/2025	2026-05-11	
12512	Daniel Rogers	upd_nlee@example.com	(288)940-4164x53130	Unit 5034 Box 1825 DPO AA 73143	2/12/2025	4/1/2025	2026-05-11	
12515	John Ray	upd_felicianchols@example.com	001-242-210-6246x4265	99379 Jones Harbor South Alexanderport, CO 21816	3/17/2025	4/1/2025	2026-05-11	
12614	Denise Allen	upd_gallegosjoshua@example.org	+1-328-582-2275x7303	8965 Horton Mews Apt. 777 Chavezmouth, ID 98416	2/8/2025	4/1/2025	2026-05-11	

3. Inserting new data of customers who have changed whit setting start_date to today, end_date to null and is_current to TRUE:

```
--new records for customers with changed data--
INSERT INTO TABLE customer_scd2_new
SELECT
    upd.id,
    upd.name,
    upd.email,
    upd.phone,
    upd.address,
    upd.join_date,
    current_date() AS start_date,
    NULL AS end_date,
    true AS is_current
FROM customer_updates upd
JOIN customer_scd2 old
ON upd.id = old.id
WHERE old.is_current = true
AND (
    old.name <> upd.name
    OR old.email <> upd.email
    OR old.phone <> upd.phone
    OR old.address <> upd.address
);
```

AZ id	AZ name	AZ email	AZ phone	AZ address	AZ join_date	start_date	end_date	is_c
12364	Larry Hess	meyerjennifer@example.com	001-626-971-5766	556 Schmidt Gateway Apt. 111 Kaneton, NM 59573	2025-02-11	2026-05-11	[NULL]	
12402	John Robinson	sandrance@example.net	919.380.4487x128	USNV Wagner FPO AA 72667	2025-01-30	2026-05-11	[NULL]	
12424	Trevor Brewer	terri28@example.com	(395)241-5789x56943	56957 Jeanne Cove Suite 783 Krystalfort, PW 47290	2025-02-07	2026-05-11	[NULL]	
12455	Chad Hubbard	tracy28@example.net	001-988-255-3862x492	5145 Joshua Plain Apt. 820 Michaelland, AK 78543	2025-02-03	2026-05-11	[NULL]	
12461	Christina Lam	samuel91@example.net	+1-657-729-1463x3734	80779 Duke Drive South Franklin, IL 66035	2025-03-24	2026-05-11	[NULL]	
12473	Brian Lucas	alexanderellis@example.org	-5790	32286 Christopher Stravenue West Lauren, MH 63912	2025-03-29	2026-05-11	[NULL]	
12512	Daniel Rogers	nlee@example.com	(288)940-4164x53130	Unit 5034 Box 1825 DPO AA 73143	2025-02-12	2026-05-11	[NULL]	
12515	John Ray	felicianichols@example.com	001-242-210-6246x4265	99279 Jones Harbor South Alexanderport, CO 21816	2025-03-17	2026-05-11	[NULL]	
12614	Denise Allen	gallegosjoshua@example.org	+1-328-582-2275x7303	8965 Horton Mews Apt. 777 Chavezmouth, ID 98416	2025-02-08	2026-05-11	[NULL]	

4. Inserting data of new customers with specifying start_date to today, end_date to null and is_current to TRUE:

```
--new customers--
INSERT INTO TABLE customer_scd2_new
SELECT
    upd.id,
    upd.name,
    upd.email,
    upd.phone,
    upd.address,
    upd.join_date,
    current_date() AS start_date,
    NULL AS end_date,
    true AS is_current
FROM customer_updates upd
LEFT JOIN customer_scd2 old
ON upd.id = old.id
WHERE old.id IS NULL;
```

AZ id	AZ name	AZ email	AZ phone	AZ address	AZ join_date	start_date	end_date	is_cu
12346	Debra Wells	michaelkerr@example.net	3317035462	USNV Olsen FPO AA 43254	2025-02-10	2026-05-11	[NULL]	
12347	Michael Grimes	rchandler@example.net	243.521.3090	36923 Bowers Gateway Suite 027 New Kristi, MP 4431	2025-03-14	2026-05-11	[NULL]	
12348	Jennifer Cuevas	brandonbrown@example.com	+1-863-584-5200x273	095 Stevenson Lodge New Brettville, LA 67557	2025-01-03	2026-05-11	[NULL]	
12349	Anthony Harvey	dgonzales@example.net	771-305-3764x583	549 Taylor Mews Baileyburgh, NJ 59083	2025-04-01	2026-05-11	[NULL]	
12350	Thomas Martin	longpatricia@example.net	558-648-9504x3880	294 Patricia Light Suite 377 Marymouth, GA 49741	2025-02-07	2026-05-11	[NULL]	
12352	James Bush	dennis40@example.net	+1-650-456-4394x097	44332 Janet Cliffs Apt. 178 Johnsonfort, WA 55056	2025-03-02	2026-05-11	[NULL]	
12353	William Greene	floreschristina@example.net	001-609-772-9593x4809	5563 Brian Drives New Charlotteland, NY 81878	2025-03-15	2026-05-11	[NULL]	
12354	Emily Tanner MD	anna45@example.com	(886)301-5618	020 Willie Springs Apt. 266 Sueberg, GA 65464	2025-01-29	2026-05-11	[NULL]	
12355	James Acevedo	josephbrown@example.com	339-695-7693	1347 Taylor Station Suite 362 South Scottport, NH 326	2025-02-24	2026-05-11	[NULL]	

5. Overwrite the data from to customer_scd2

The final data in the customer_SCD2:

```
INSERT OVERWRITE TABLE customer_scd2
SELECT * FROM customer_scd2_new;

TRUNCATE TABLE customer_scd2_new;

SELECT * FROM customer_scd2;
```

Results 1 X

SELECT * FROM customer_scd2_new; Enter a SQL expression to filter results (use Ctrl+Space)

		AZ_email	AZ_phone	AZ_address	AZ_join_date	AZ_start_date	AZ_end_date	AZ_is_current
120	ler	upd_julie01@example.org	+1-800-305-0916x717	90729 Wright Tunnel Michaelbury, IL 53815	1/3/2025	4/1/2025	2026-05-11	FALSE
121	rdner	upd_henry43@example.org	549-644-2653x762	PSC 5821, Box 4049 APO AA 07029	1/27/2025	4/1/2025	2026-05-11	FALSE
122	n	upd_cmatis@example.org	922-743-9703x51860	8621 Joshua Ridge Suite 963 Elizabethbury, MS 08226	2/19/2025	4/1/2025	2026-05-11	FALSE
123		upd_trasmussen@example.com	9806921955	0757 Brian Lane Apt. 857 South Anthonyland, CO 90828	3/27/2025	4/1/2025	2026-05-11	FALSE
124	son	upd_mcdonaldjesse@example.com	(869)921-8538	07525 Espinoza Rest Eileenland, PW 05450	1/19/2025	4/1/2025	2026-05-11	FALSE
125	za	upd_gordonkeith@example.org	612-247-6675x190	4551 Davis Forest Apt. 998 North Zachary, CA 84515	3/1/2025	4/1/2025	2026-05-11	FALSE
126	gham	upd_debbiesimmons@example.com	529-621-2207	5241 Leah Shoals Amandatown, KY 60131	1/24/2025	4/1/2025	2026-05-11	FALSE
127	s	upd_jeremy38@example.net	431-307-6479x036	329 Diana Lights Suite 064 Tuckerborough, KY 76020	1/11/2025	4/1/2025	2026-05-11	FALSE
128	e	upd_mwoods@example.net	264-334-0342	271 Smith Path Apt. 239 Port Sheilamouth, NV 07687	3/21/2025	4/1/2025	2026-05-11	FALSE
129	van	upd_rspencer@example.com	651-256-0421	43013 Smith Isle Apt. 153 Oliverview, SC 12524	3/27/2025	4/1/2025	2026-05-11	FALSE
130	h	upd_lawrence81@example.org	+1-467-738-7189x310	76206 Steele Circle Lake Nicholas, WA 46561	1/27/2025	4/1/2025	2026-05-11	FALSE
131		meyerjennifer@example.com	001-626-971-5766	556 Schmidt Gateway Apt. 111 Kaneton, NM 59573	2025-02-11	2026-05-11		TRUE
132	on	sandrarice@example.net	919-380-4487x128	USNV Wagner FPO AA 72667	2025-01-30	2026-05-11		TRUE
133	er	terri28@example.com	(395)241-5789x56943	56957 Jeanne Cove Suite 783 Krystallfort, PW 47290	2025-02-07	2026-05-11		TRUE
134	ird	tracy28@example.net	001-988-255-3862x492	5145 Joshua Plain Apt. 820 Michaeland, AK 78543	2025-02-03	2026-05-11		TRUE
135	n	samuel91@example.net	+1-657-729-1463x3734	80779 Duke Drive South Franklin, IL 66035	2025-03-24	2026-05-11		TRUE
136		alexanderellis@example.org	-5790	32286 Christopher Stravenue West Lauren, MH 63912	2025-03-29	2026-05-11		TRUE
137	rs	nlee@example.com	(288)940-4164x53130	Unit 5034 Box 1825 DPO AA 73143	2025-02-12	2026-05-11		TRUE
138		felicianichols@example.com	001-242-210-6246x4265	99279 Jones Harbor South Alexanderport, CO 21816	2025-03-17	2026-05-11		TRUE
139	s	gallegosjoshua@example.org	+1-328-582-2275x7303	8965 Horton Mews Apt. 777 Chavezmouth, ID 98416	2025-02-08	2026-05-11		TRUE
140	DVM	fpope@example.com	001-349-611-7306x9555	732 Cooper Locks East Carlos, GU 73839	2025-03-16	2026-05-11		TRUE
141	hony	zshahia97@example.net	775-370-6374	06904 Timothy Plaza Suite 020 North Andersonland, LA 66070	2025-02-11	2026-05-11		TRUE